

ESBD Research Report

Correlated Randomized Gamma Variables and Their Application to CEV
Processes

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Contents

1	Introduction	2
2	Correlated Randomized Gamma Distributions	2
2.1	Objective	2
2.2	Simulation methods	2
2.2.1	Method 1: Exponential-sum construction	2
2.2.2	Method 2: Inverse-CDF construction	3
2.3	Correlation derivations	3
2.3.1	Common randomizer: $N = M$	3
2.3.2	Independent randomizers: N and M	5
2.4	Simulation results	5
2.4.1	Case 1: Common Poisson randomizer $N = M$	5
2.4.2	Case 2: Independent Poisson randomizers	7
3	Application to Correlated CEV Processes	8
3.1	Objective	8
3.2	CEV to SQB transformation	8
3.3	Simulation algorithms	9
3.3.1	Case 1: Simulation without an explicit absorption mechanism	9
3.3.2	Case 2: Simulation with absorption	10
3.4	One-step SQB correlation derivations	11
3.4.1	Without absorption	11
3.4.2	With absorption	11
3.5	CEV/SQB simulation results	12
3.5.1	Verification of the one-step SQB correlation without absorption	13
3.5.2	CEV correlation through time	13
3.5.3	One-step SQB correlation with absorption	14
4	Conclusion	15

1 Introduction

This report studies the construction of correlated randomized Gamma random variables and their use in exact simulation of correlated CEV processes through a squared Bessel (SQB) representation. We begin by comparing two simulation methods for randomized Gamma variables and derive their effective correlation under two randomizer structures. Then, we apply the randomized Gamma construction to the CEV to SQB transformation, with particular emphasis on the negative correlations that can be achieved between two asset price processes.

Throughout the report, the notation $\text{Gamma}(a, b)$ denotes a Gamma distribution with shape a and rate b . Consequently, $\text{Exp}(b)$ denotes an exponential distribution with rate b .

2 Correlated Randomized Gamma Distributions

2.1 Objective

We compare two methods for generating correlated randomized Gamma random variables X and Y . Two randomizer structures are considered:

Case 1: a common Poisson randomizer, $N = M$, with $N \sim \text{Pois}(\lambda)$;

Case 2: independent Poisson randomizers, $N, M \stackrel{\text{iid}}{\sim} \text{Pois}(\lambda)$.

For each case, we compare the simulated value of $\text{Corr}(X, Y)$ with its theoretical counterpart and compare the computational cost of the two simulation methods.

2.2 Simulation methods

2.2.1 Method 1: Exponential-sum construction

Let $0 < \alpha_0 < 1$ and let $\beta > 0$ denote the Gamma rate. The first method uses the decomposition of a Gamma random variable into a base Gamma term and a random number of exponential increments.

1. Simulate independent base variables

$$G_{0,X}, G_{0,Y} \stackrel{\text{iid}}{\sim} \text{Gamma}(\alpha_0, \beta).$$

2. Generate the randomizers N and M . In Case 1, simulate $N \sim \text{Pois}(\lambda)$ and set $M = N$. In Case 2, simulate $N, M \stackrel{\text{iid}}{\sim} \text{Pois}(\lambda)$.

3. Generate a bivariate standard normal vector using a Gaussian copula,

$$\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho_Z \\ \rho_Z & 1 \end{pmatrix}\right), \quad -1 < \rho_Z < 1,$$

and set

$$U_1 = \Phi(Z_1), \quad U_2 = \Phi(Z_2).$$

Marginally, $U_1, U_2 \sim \text{Unif}(0, 1)$.

4. Construct the correlated exponential pairs $E_X, E_Y \sim \text{Exp}(\beta)$,

$$E_X = -\frac{\log U_1}{\beta}, \quad E_Y = -\frac{\log U_2}{\beta},$$

The first $\min(N, M)$ exponential terms are generated as correlated pairs, while any unmatched terms are generated independently.

5. Construct

$$X = G_{0,X} + \sum_{k=1}^N E_{X,k}, \quad Y = G_{0,Y} + \sum_{k=1}^M E_{Y,k}. \quad (1)$$

Conditional on the randomizers,

$$(X \mid N) \sim \text{Gamma}(\alpha_0 + N, \beta), \quad (Y \mid M) \sim \text{Gamma}(\alpha_0 + M, \beta).$$

2.2.2 Method 2: Inverse-CDF construction

The second method generates the same conditional Gamma marginals directly through the inverse Gamma CDF.

1. Generate N and M as in Method 1.
2. Generate (U_1, U_2) from the same Gaussian copula.
3. Set

$$X = F_{\text{Gamma}(\alpha_0 + N, \beta)}^{-1}(U_1), \quad Y = F_{\text{Gamma}(\alpha_0 + M, \beta)}^{-1}(U_2).$$

Because MATLAB's `gaminv` function uses a scale parameter rather than a rate parameter, the implementation corresponding to the notation in this report is

$$\begin{aligned} X &= \text{gaminv}(U_1, \alpha_0 + N, 1/\beta), \\ Y &= \text{gaminv}(U_2, \alpha_0 + M, 1/\beta). \end{aligned}$$

2.3 Correlation derivations

Let

$$\rho_E = \text{Corr}(E_X, E_Y).$$

Since $E_X, E_Y \sim \text{Exp}(\beta)$,

$$\text{Var}(E_X) = \text{Var}(E_Y) = \frac{1}{\beta^2}, \quad \text{Cov}(E_X, E_Y) = \frac{\rho_E}{\beta^2}.$$

2.3.1 Common randomizer: $N = M$

Suppose N is a nonnegative integer-valued random variable with finite mean and variance and that X, Y are constructed as in equation (1) with $M = N$.

By the law of total covariance,

$$\text{Cov}(X, Y) = \mathbb{E}[\text{Cov}(X, Y \mid N)] + \text{Cov}(\mathbb{E}[X \mid N], \mathbb{E}[Y \mid N]).$$

Conditional on N , the only covariance comes from the N paired exponentials, so

$$\text{Cov}(X, Y \mid N) = N \frac{\rho_E}{\beta^2}.$$

Therefore,

$$\mathbb{E}[\text{Cov}(X, Y \mid N)] = \frac{\rho_E \mathbb{E}[N]}{\beta^2}.$$

Also,

$$\mathbb{E}[X \mid N] = \mathbb{E}[Y \mid N] = \frac{\alpha_0 + N}{\beta},$$

which gives

$$\text{Cov}(\mathbb{E}[X \mid N], \mathbb{E}[Y \mid N]) = \frac{\text{Var}(N)}{\beta^2}.$$

Hence,

$$\text{Cov}(X, Y) = \frac{\rho_E \mathbb{E}[N] + \text{Var}(N)}{\beta^2}.$$

Using the law of total variance,

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid N)] + \text{Var}(\mathbb{E}[X \mid N])$$

Now,

$$\text{Var}(X \mid N) = \frac{\alpha_0 + N}{\beta^2}.$$

Thus,

$$\mathbb{E}[\text{Var}(X \mid N)] = \frac{\alpha_0 + \mathbb{E}[N]}{\beta^2},$$

and

$$\text{Var}(\mathbb{E}[X \mid N]) = \frac{\text{Var}(N)}{\beta^2}$$

Hence,

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[\text{Var}(X \mid N)] + \text{Var}(\mathbb{E}[X \mid N]) \\ &= \frac{\alpha_0 + \mathbb{E}[N]}{\beta^2} + \frac{\text{Var}(N)}{\beta^2} \\ &= \frac{\alpha_0 + \mathbb{E}[N] + \text{Var}(N)}{\beta^2}. \end{aligned}$$

The same expression holds for $\text{Var}(Y)$. Therefore,

$$\boxed{\text{Corr}(X, Y) = \frac{\rho_E \mathbb{E}[N] + \text{Var}(N)}{\alpha_0 + \mathbb{E}[N] + \text{Var}(N)}}. \quad (2)$$

For $N \sim \text{Pois}(\lambda)$, $\mathbb{E}[N] = \text{Var}(N) = \lambda$, so equation (2) becomes

$$\boxed{\text{Corr}(X, Y) = \frac{\lambda(1 + \rho_E)}{\alpha_0 + 2\lambda}}. \quad (3)$$

2.3.2 Independent randomizers: N and M

Now suppose N and M are independent nonnegative integer-valued random variables. Conditional on (N, M) , only the first $\min(N, M)$ exponentials contribute to the covariance. Thus,

$$\text{Cov}(X, Y \mid N, M) = \min(N, M) \frac{\rho_E}{\beta^2}.$$

Moreover,

$$\text{Cov}(\mathbb{E}[X \mid N, M], \mathbb{E}[Y \mid N, M]) = \frac{\text{Cov}(N, M)}{\beta^2} = 0.$$

Therefore,

$$\text{Cov}(X, Y) = \mathbb{E}[\text{Cov}(X, Y \mid N)] + \text{Cov}(\mathbb{E}[X \mid N], \mathbb{E}[Y \mid N]) = \frac{\rho_E \mathbb{E}[\min(N, M)]}{\beta^2}.$$

From the case with equal randomizers, we can identify the marginal variances as

$$\text{Var}(X) = \frac{\alpha_0 + \mathbb{E}[N] + \text{Var}(N)}{\beta^2}, \quad \text{Var}(Y) = \frac{\alpha_0 + \mathbb{E}[M] + \text{Var}(M)}{\beta^2}.$$

Thus,

$$\boxed{\text{Corr}(X, Y) = \frac{\rho_E \mathbb{E}[\min(N, M)]}{\sqrt{(\alpha_0 + \mathbb{E}[N] + \text{Var}(N))(\alpha_0 + \mathbb{E}[M] + \text{Var}(M))}}}. \quad (4)$$

For independent $N, M \stackrel{\text{iid}}{\sim} \text{Pois}(\lambda)$, this simplifies to

$$\boxed{\text{Corr}(X, Y) = \frac{\rho_E \mathbb{E}[\min(N, M)]}{\alpha_0 + 2\lambda}}. \quad (5)$$

2.4 Simulation results

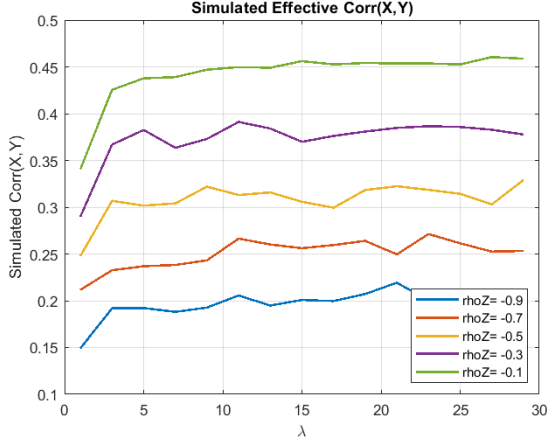
For the randomized-Gamma experiments, we generated 10,000 simulations using the parameter values

$$\alpha_0 = \frac{1}{2}, \quad \beta = 1.$$

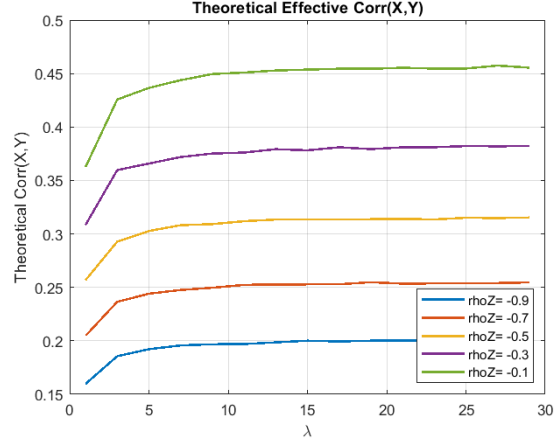
The Poisson parameter λ and Gaussian-copula parameter ρ_Z were varied. For each pair (λ, ρ_Z) , the effective correlation $\text{Corr}(X, Y)$ was estimated and compared with the theoretical expression derived in equation (3) and equation (5). Computation times for Methods 1 and 2 were also recorded.

2.4.1 Case 1: Common Poisson randomizer $N = M$

Figure 1 compares the simulated and theoretical effective correlations. The two sets of curves are closely aligned, providing numerical validation of equation (3). The correlation tends to increase as λ increases, and larger magnitudes of ρ_Z lead to smaller effective correlation results.



(a) Simulated effective correlation.



(b) Theoretical effective correlation.

Figure 1: Effective correlation for the common-randomizer case $N = M$.

Next, we investigate the limiting value of the correlation using the lower Fréchet bound. At the lower Fréchet bound, $\rho_Z \rightarrow -1$, so $U_2 = 1 - U_1$. When $\beta = 1$,

$$E_X = -\log U, \quad E_Y = -\log(1 - U),$$

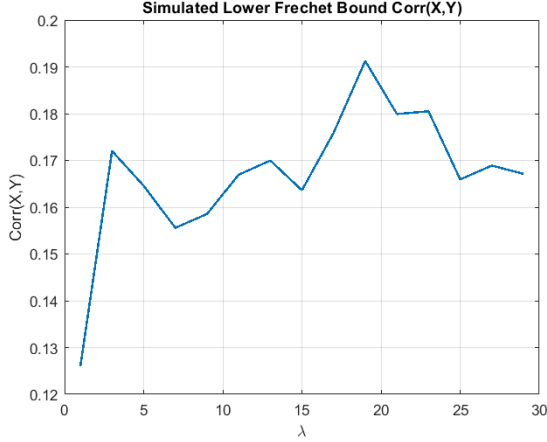
and

$$\begin{aligned} \rho_E &= \text{Corr}(E_X, E_Y) \\ &= \text{Cov}(E_X, E_Y) \\ &= \mathbb{E}[E_X E_Y] - \mathbb{E}[E_X] \mathbb{E}[E_Y] \\ &= \int_0^1 \log(u) \log(1 - u) du - 1 \\ &= 1 - \frac{\pi^2}{6} \approx -0.644934. \end{aligned}$$

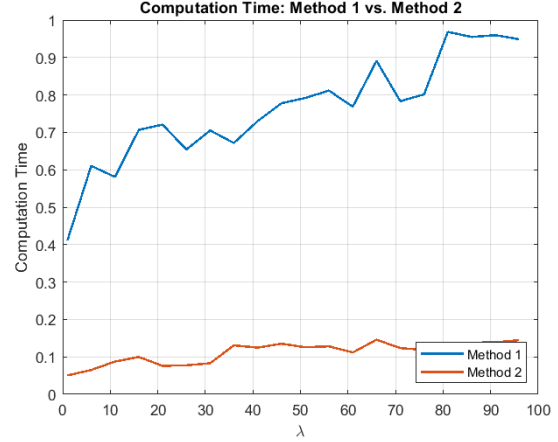
Substituting this into equation (3) with $\alpha_0 = 1/2$ gives

$$\text{Corr}(X, Y) = \frac{\lambda(2 - \pi^2/6)}{1/2 + 2\lambda} \longrightarrow \frac{2 - \pi^2/6}{2} \approx 0.177533 \text{ as } \lambda \rightarrow \infty$$

The simulated lower-bound correlation in figure 2 approaches the limit found above as λ continues to increase.



(a) Simulated lower Fréchet-bound correlation.



(b) Computation time for Methods 1 and 2.

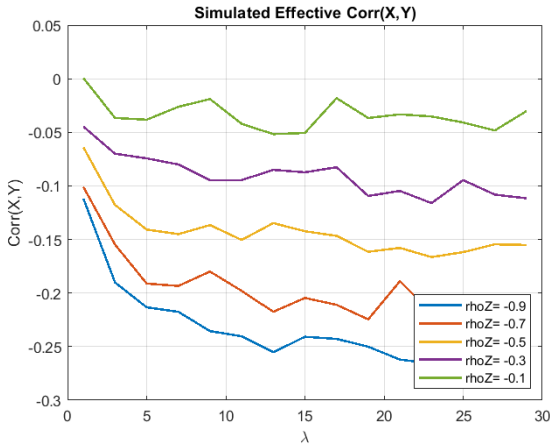
Figure 2: Lower-bound behavior and computational efficiency for the common-randomizer case.

The timing comparison in figure 2b shows that Method 2 is substantially faster over the tested range of λ . Method 1 becomes increasingly expensive because a larger value of λ increases the expected number of exponential increments that must be generated and summed.

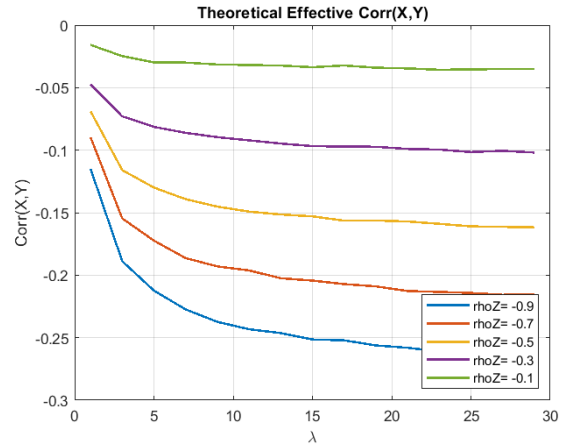
2.4.2 Case 2: Independent Poisson randomizers

Figure 3 compares simulated and theoretical correlations when N and M are independent. Again, the simulation closely follows the theoretical curves, supporting equation (5). The effective correlations have smaller magnitudes than in the common-randomizer case because the covariance contribution from $\text{Cov}(N, M)$ disappears. In addition,

$$\mathbb{E}[\min(N, M)] \leq \mathbb{E}[N] = \lambda.$$



(a) Simulated effective correlation.



(b) Theoretical effective correlation.

Figure 3: Effective correlation for independent Poisson randomizers.

For large λ , $\mathbb{E}[\min(N, M)] \approx \lambda$. Thus, $\mathbb{E}[\min(N, M)]/\lambda \rightarrow 1$. Hence, from equation (5),

$$\text{Corr}(X, Y) \approx \frac{\rho_E}{2}.$$

At the lower Fréchet bound,

$$\text{Corr}(X, Y) \rightarrow \frac{1 - \pi^2/6}{2} \approx -0.322467.$$

The simulation in figure 4 is consistent with convergence toward this negative limiting value found above.

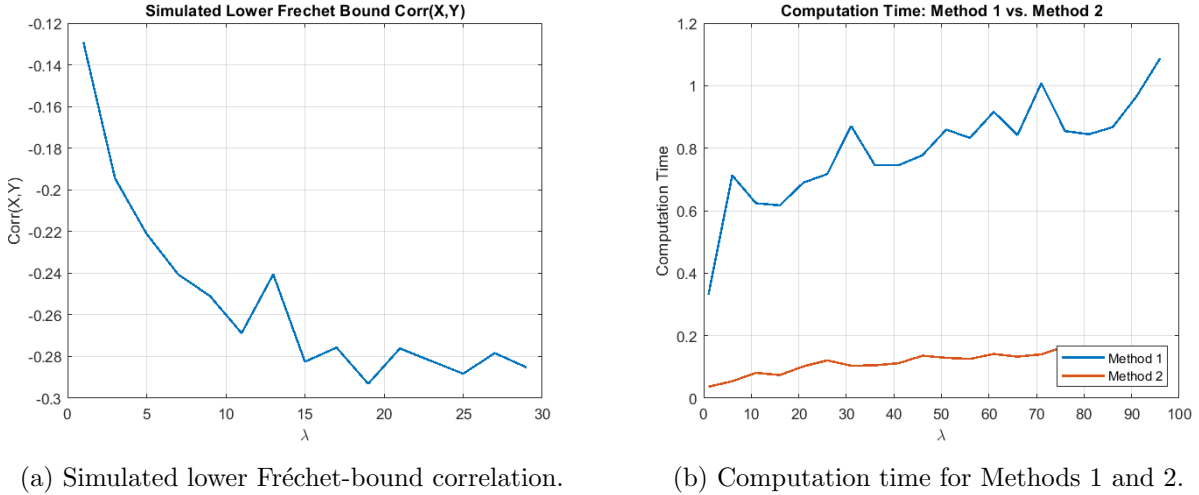


Figure 4: Lower-bound behavior and computational efficiency for independent randomizers.

As in the common-randomizer case, Method 2 is faster over the tested parameter range, and as λ increases, the computational time between the two methods becomes more pronounced.

3 Application to Correlated CEV Processes

3.1 Objective

We next investigate the correlation that can be achieved between two asset prices following CEV processes when exact simulation is performed through corresponding SQB processes, with emphasis on negative dependence. Each CEV process $S_j(t)$, $j = 1, 2$, is transformed into an SQB process $X_j(t)$, whose one-step transition can be represented as a randomized Gamma variable. Dependence is then introduced through the randomized-Gamma construction before mapping the simulated SQB values back to CEV prices.

3.2 CEV to SQB transformation

Consider the CEV diffusion

$$dS_t = \mu S_t dt + \delta S_t^{\beta+1} dW_t. \quad (6)$$

For $\beta < 0$, define

$$X_t = \frac{S_t^{-2\beta}}{\delta^2 \beta^2}. \quad (7)$$

Under this transformation, the SQB dimension parameter is

$$\nu = 2 + \frac{1}{\beta}, \quad \alpha_0 = \frac{\nu}{2} = 1 + \frac{1}{2\beta}.$$

For the case with absorption, $\beta < -1/2$ and consequently, $0 < \alpha_0 < 1$.

After simulating the SQB process, the corresponding CEV value is recovered using

$$S_t = (\delta^2 \beta^2 X_t)^{-1/(2\beta)}. \quad (8)$$

This representation allows dependence to be introduced in the exact randomized Gamma transition rather than through an Euler discretization of the original CEV SDE given by equation (6).

3.3 Simulation algorithms

The numerical study assumes $\mu_j = 0$ for $j = 1, 2$, where j denotes which asset we are working with.

3.3.1 Case 1: Simulation without an explicit absorption mechanism

1. Specify $(S_{j,0}, \mu_j, \sigma_{j,0}, \beta_j)$, $j = 1, 2$, and calculate

$$\delta_j = \frac{\sigma_{j,0}}{S_{j,0}^{\beta_j}}.$$

2. Transform the initial CEV values to SQB values using equation (7).

3. At time step t_i , calculate

$$\lambda_{j,t_i} = \frac{X_{j,t_i}}{2\Delta t}$$

and independently sample

$$N_{j,t_i} \sim \text{Pois}(\lambda_{j,t_i}).$$

4. Independently sample the base Gamma terms

$$G_{j,0} \sim \text{Gamma}\left(\alpha_{0,j}, \frac{1}{2\Delta t}\right), \quad \alpha_{0,j} = 1 + \frac{1}{2\beta_j}.$$

5. Generate independent $Z_1, Z_2 \sim N(0, 1)$ and define

$$\tilde{Z} = \rho Z_1 + \sqrt{1 - \rho^2} Z_2.$$

Set

$$U_1 = \Phi(Z_1), \quad U_2 = \Phi(\tilde{Z}),$$

and construct exponential increments

$$E_1 = -2\Delta t \log U_1, \quad E_2 = -2\Delta t \log U_2.$$

Such that $E_1, E_2 \sim \text{Exp}(1/(2\Delta t))$, where $1/(2\Delta t)$ is the rate parameter. The first

$$m_{t_i} = \min(N_{1,t_i}, N_{2,t_i})$$

increments are generated as correlated pairs.

6. Generate any unmatched exponential increments independently.
7. Set

$$X_{j,t_{i+1}} \mid X_{j,t_i} = G_{j,0} + \sum_{k=1}^{N_{j,t_i}} E_{j,k}. \quad (9)$$

8. Transform $X_{j,i+1}$ back to $S_{j,i+1}$ using equation (8).
9. Repeat steps 3-8 until the terminal time T is reached.

3.3.2 Case 2: Simulation with absorption

For the absorbing construction, define

$$\mu_{j,\text{SQB}} = \frac{1}{2\beta_j}, \quad \eta_j = |\mu_{j,\text{SQB}}|.$$

The simulation proceeds as follows.

1. Specify the CEV parameters, calculate δ_j , and map the initial values to SQB states as above.
2. At each time step t_i , check whether process j has already been absorbed. If so, set $X_{j,i+1} = S_{j,i+1} = 0$ and do not generate further random variables for that process.
3. For each surviving process, calculate

$$\lambda_{j,t_i} = \frac{X_{j,t_i}}{2\Delta t}.$$

4. Compute the conditional absorption and survival probabilities

$$p_{a,j,t_i} = \frac{\Gamma(\eta_j, \lambda_{j,t_i})}{\Gamma(\eta_j)}, \quad p_{s,j,t_i} = \frac{\gamma(\eta_j, \lambda_{j,t_i})}{\Gamma(\eta_j)} = 1 - p_{a,j,t_i}.$$

5. Independently sample

$$B_{j,t_i} \sim \text{Bernoulli}(p_{s,j,t_i}).$$

Conditional on the current SQB states, the two survival indicators are assumed independent.

6. If $B_{j,t_i} = 0$, the process is absorbed during the current time step t_i . Set $X_{j,t_{i+1}} = S_{j,t_{i+1}} = 0$, and keep both processes at 0 for all subsequent time steps.
7. Conditional on $B_{j,t_i} = 1$, independently sample the shifted-Poisson randomizer

$$N_{j,t_i} \sim I\Gamma(\eta_j, \lambda_{j,t_i}).$$

8. Independently sample

$$G_{j,0} \sim \text{Gamma}\left(1, \frac{1}{2\Delta t}\right).$$

9. Generate correlated exponential increments through the Gaussian copula exactly as in the non-absorbing construction, pairing the first $\min(N_{1,t_i}, N_{2,t_i})$ increments.
10. Conditional on survival, set

$$X_{j,t_{i+1}} \mid X_{j,t_i} = G_{j,0} + \sum_{k=1}^{N_{j,t_i}} E_{j,k} \quad (10)$$

and transform back to CEV using equation (8).

11. Repeat steps 3-10 until T is reached.

3.4 One-step SQB correlation derivations

3.4.1 Without absorption

For the one-step transition in equation (9), let

$$\text{Corr}(E_{1,t_i}, E_{2,t_i}) = \rho_E.$$

Given current states (X_{1,t_i}, X_{2,t_i}) , write

$$\mu_{N_j} = \mathbb{E}[N_{j,t_i} \mid X_{1,t_i}, X_{2,t_i}], \quad \sigma_{N_j}^2 = \text{Var}(N_{j,t_i} \mid X_{1,t_i}, X_{2,t_i}).$$

Because an exponential random variable with rate $1/(2\Delta t)$ has variance $4(\Delta t)^2$, and using the results from the generalized randomized gamma derivation.

$$\begin{aligned} & \text{Cov}(X_{1,t_{i+1}}, X_{2,t_{i+1}} \mid X_{1,t_i}, X_{2,t_i}) \\ &= 4(\Delta t)^2 \left[\rho_E \mathbb{E}[\min(N_{1,t_i}, N_{2,t_i}) \mid X_{1,t_i}, X_{2,t_i}] + \text{Cov}(N_{1,t_i}, N_{2,t_i} \mid X_{1,t_i}, X_{2,t_i}) \right]. \end{aligned}$$

The conditional marginal variances are

$$\text{Var}(X_{j,t_{i+1}} \mid X_{1,t_i}, X_{2,t_i}) = 4(\Delta t)^2 \left(\alpha_{0,j} + \mu_{N_j} + \sigma_{N_j}^2 \right).$$

Therefore,

$$\boxed{\text{Corr}(X_{1,t_{i+1}}, X_{2,t_{i+1}} \mid X_{1,t_i}, X_{2,t_i}) = \frac{\rho_E \mathbb{E}[\min(N_{1,t_i}, N_{2,t_i}) \mid X_{1,t_i}, X_{2,t_i}] + \text{Cov}(N_{1,t_i}, N_{2,t_i} \mid X_{1,t_i}, X_{2,t_i})}{\sqrt{(\alpha_{0,1} + \mu_{N_1} + \sigma_{N_1}^2)(\alpha_{0,2} + \mu_{N_2} + \sigma_{N_2}^2)}}} \quad (11)$$

If the randomizers are conditionally independent, the covariance term in the numerator is zero.

3.4.2 With absorption

Let B_{j,t_i} be the survival indicator over $[t_i, t_{i+1}]$:

$$B_{j,t_i} = \begin{cases} 1, & \text{process } j \text{ survives the time step,} \\ 0, & \text{process } j \text{ is absorbed.} \end{cases}$$

Define

$$p_{j,t_i} = \mathbb{P}(B_{j,i} = 1 \mid X_{j,i}).$$

Assume the survival indicators are conditionally independent across the two processes and are conditionally independent of the randomized-Gamma construction. Then

$$\mathbb{E}[B_{1,t_i} B_{2,t_i} \mid X_{1,t_i}, X_{2,t_i}] = p_{1,t_i} p_{2,t_i}.$$

Conditional on survival,

$$X_{j,t_{i+1}} \mid X_{j,t_i}, N_{j,t_i}, B_{j,t_i} \sim \text{Gamma}\left(1 + N_{j,t_i}, \frac{1}{2\Delta t}\right).$$

Hence, from the generalized randomized gamma derivation,

$$\mathbb{E}[X_{j,t_{i+1}} \mid X_{j,t_i}, B_{j,t_i}] = 2\Delta t (1 + \mathbb{E}[N_{j,t_i} \mid X_{j,t_i}]),$$

and

$$\text{Var}(X_{j,t_{i+1}} \mid X_{j,t_i}, B_{j,t_i}) = 4(\Delta t)^2 (1 + \mathbb{E}[N_{j,t_i} \mid X_{j,t_i}] + \text{Var}(N_{j,t_i} \mid X_{j,t_i})).$$

By the laws of total expectation and total variance,

$$\mathbb{E}[X_{j,t_{i+1}} \mid X_{j,t_i}] = 2\Delta t p_{j,t_i} (1 + \mathbb{E}[N_{j,t_i} \mid X_{j,t_i}]), \quad (12)$$

$$\begin{aligned} \text{Var}(X_{j,t_{i+1}} \mid X_{j,t_i}) &= 4(\Delta t)^2 \left[p_{j,t_i} (1 + \mathbb{E}[N_{j,t_i} \mid X_{j,t_i}] + \text{Var}(N_{j,t_i} \mid X_{j,t_i})) \right. \\ &\quad \left. + p_{j,t_i} (1 - p_{j,t_i}) (1 + \mathbb{E}[N_{j,t_i} \mid X_{j,t_i}])^2 \right]. \end{aligned} \quad (13)$$

Conditional on joint survival and referencing the generalized randomized gamma derivation, the covariance is

$$\begin{aligned} \text{Cov}(X_{1,t_{i+1}}, X_{2,t_{i+1}} \mid X_{1,t_i}, X_{2,t_i}, B_{1,t_i}, B_{2,t_i}) \\ = 4(\Delta t)^2 \left[\rho_E \mathbb{E}[\min(N_{1,t_i}, N_{2,t_i}) \mid X_{1,t_i}, X_{2,t_i}] + \text{Cov}(N_{1,t_i}, N_{2,t_i} \mid X_{1,t_i}, X_{2,t_i}) \right]. \end{aligned}$$

Because the survival indicators are conditionally independent,

$$\begin{aligned} \text{Cov}(X_{1,t_{i+1}}, X_{2,t_{i+1}} \mid X_{1,t_i}, X_{2,t_i}) \\ = 4p_{1,t_i} p_{2,t_i} (\Delta t)^2 \left[\rho_E \mathbb{E}[\min(N_{1,t_i}, N_{2,t_i}) \mid X_{1,t_i}, X_{2,t_i}] + \text{Cov}(N_{1,t_i}, N_{2,t_i} \mid X_{1,t_i}, X_{2,t_i}) \right]. \end{aligned} \quad (14)$$

The one-step correlation follows by dividing equation (14) by the square root of the two variances in equation (13). If the randomizers are conditionally independent, the conditional covariance of the randomizers is zero.

3.5 CEV/SQB simulation results

The experiments use the parameters

$$\beta_1 = -0.6, \quad \beta_2 = -0.7, \quad S_{1,0} = 100, \quad S_{2,0} = 80,$$

with initial local volatilities

$$\sigma_{1,0} = 0.03, \quad \sigma_{2,0} = 0.04.$$

The first-step correlation calculations use 20,000 simulated paths. The study compares the simulated and theoretical one-step SQB correlation, the effect of the nonlinear inverse transformation on first-step CEV correlation, the evolution of CEV correlation through time for different numbers of time steps.

3.5.1 Verification of the one-step SQB correlation without absorption

Figure 5 compares the first-step SQB correlation with its theoretical value and shows the corresponding first-step CEV correlation. The simulated SQB correlation closely tracks the theoretical expression across the tested values of ρ . The correlation also increases with ρ .

The CEV correlation exhibits the same overall dependence pattern after applying the nonlinear inverse transformation in equation (8). The numerical values are close in this experiment, although correlation is not generally invariant under nonlinear transformations. The appropriate conclusion is therefore that the transformation has only a small effect on the observed first-step correlation for the parameter values considered.

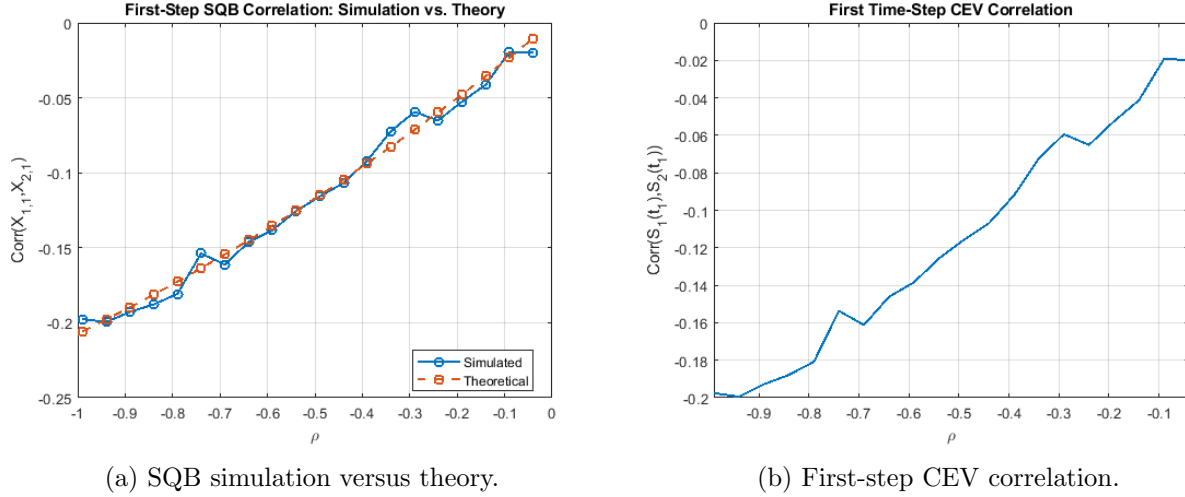


Figure 5: First-step correlation without an explicit absorption mechanism.

3.5.2 CEV correlation through time

Figure 6 shows the CEV correlation through time when the interval is represented using 1, 2, and 4 time steps at $\rho = -0.50$. The figure illustrates that the measured correlation depends on the observation time and on the number of transitions used over the horizon. The finer step simulations provide several intermediate correlation values, whereas the one-step experiment provides the correlation at time T .

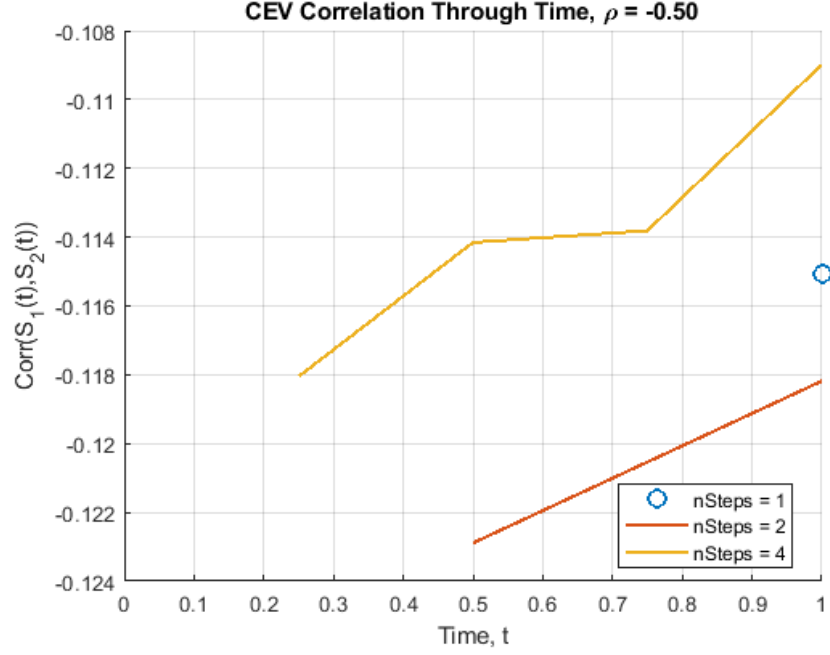


Figure 6: CEV correlation through time for different numbers of time steps, with $\rho = -0.50$.

3.5.3 One-step SQB correlation with absorption

Figure 7 compares the simulated and theoretical one-step SQB correlations when the absorption mechanism is included. As in the non-absorbing implementation, the simulated values closely follow the theoretical curve. The remaining discrepancies are small relative to the overall trend and are attributable to numerical noise occurring from the simulations.

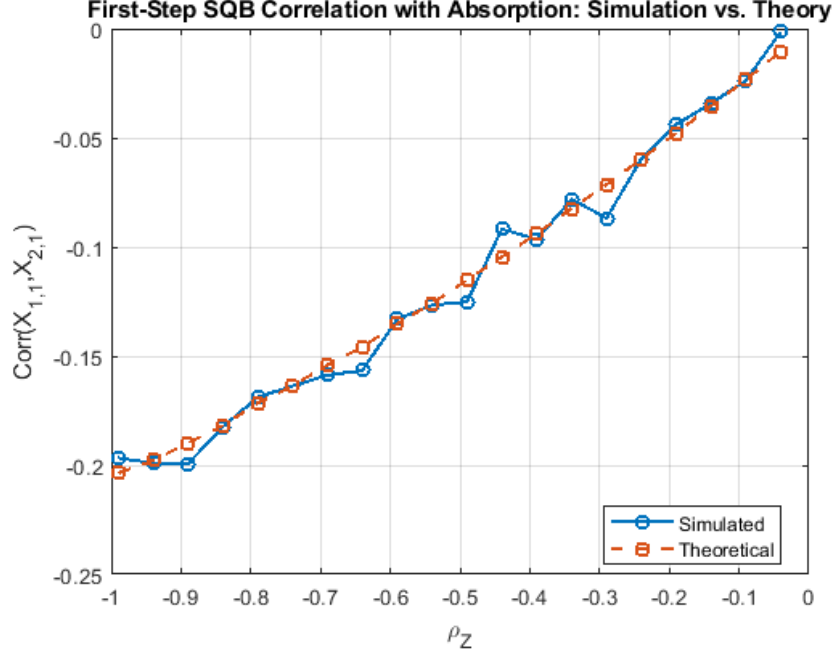


Figure 7: First-step SQB correlation with absorption: simulation versus theory.

4 Conclusion

The randomized-Gamma experiments provide strong numerical agreement between the derived correlation formulas and Monte Carlo simulation. A common randomizer creates an additional positive covariance contribution through $\text{Var}(N)$, whereas independent randomizers remove this term and permit substantially more negative effective correlations. The lower Fréchet bound experiments agree with the limiting values implied by the exponential-pair correlation $1 - \pi^2/6$.

The comparison of simulation methods also shows that the inverse-CDF construction is considerably faster than explicitly generating and summing exponential increments over the parameter ranges considered. The exponential-sum construction nevertheless remains useful because it exposes a direct mechanism for controlling dependence through paired increments and extends naturally to the SQB transition representation.

In the CEV application, the SQB simulation results agree closely with the theoretical one-step correlation formulas, both with and without the explicit absorption mechanism. Mapping the SQB processes back to CEV prices preserves the qualitative dependence pattern, while the simulations also show that the resulting correlation can vary with time and with the number of transitions over the horizon.